

IT Networking Fundamentals (Udemy)

What is Networking?

What is a computer Network?

- A computer network is defined as the interconnection of two or more computers.
- Sharing of resources such as resources, softwares and databases

Components

- Two or more computers
- Cables or connections between the computers
- A network interfacing card (NIC) on each computer
- Switches
- Software called operating system (OS)

Why network?

- Sharing
- Connectivity

Benefits

- Increased speed
- Reduced cost
- Improved security
- Centralized software management
- Electronic mail
- Flexible access

Source system

Destination system

source → Transmitter → Transmission → Receiver → Destination

1

2

3

4

5

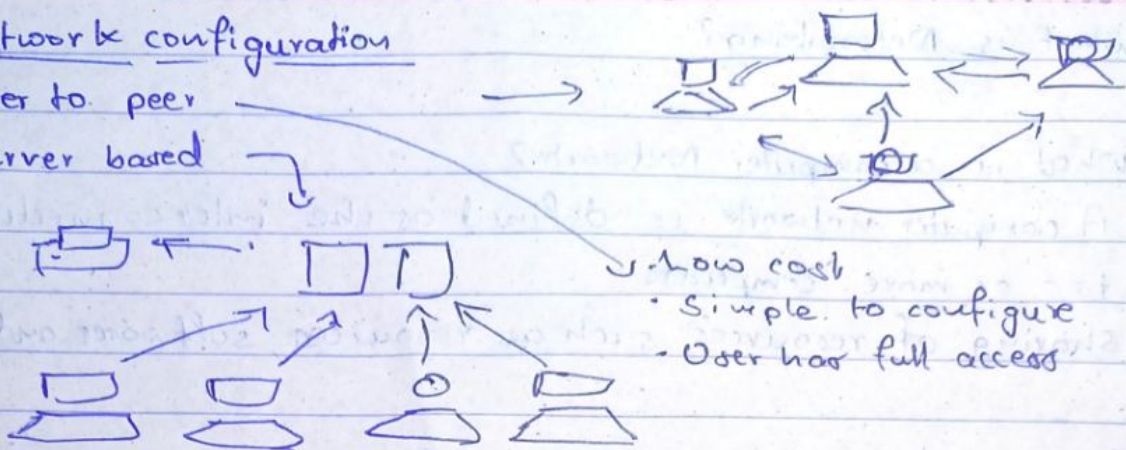
Workstation/
PC

Medium

workstation/PC

Network configuration

- Peer to peer
- Server based



Low cost

- Simple to configure
- User has full access

Server based

Peer to peer

- LAN
- Office, Factory or group of buildings.

Additional Classifications:

LAN, MAN, WAN, PAN

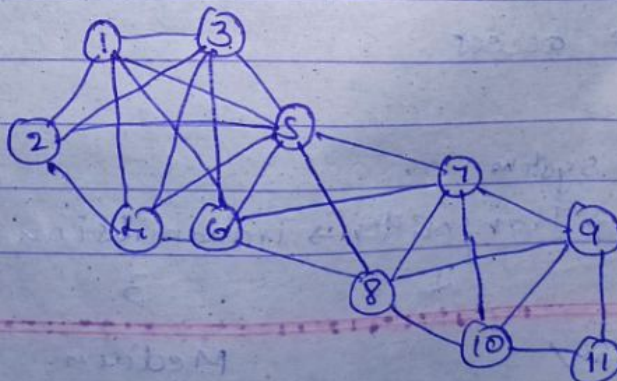
Network Topology:

- A pattern in which nodes (i.e. computers, printers, routers or other devices) are connected to a local area network or other network via links.

Types of Topologies

- Bus
- Mesh
- Ring
- Star

Mesh (Combination star & bus)



Some considerations

- Money: Bus (least expensive)
- Length of cable needed: Bus
- Future growth: Star
- Cable type: Twisted pair

IP Address

• Unique address used to identify a device

1. Private IP address

2. Public IP address

Local ip address 192.168 and 10.0

TCP / IP

• Transmission Control Protocol.

Class A address

- Assigned to ~~for~~ very large ^{no. of} hosts.

Class B

- Medium-sized to large sized networks.

Class C

- Small networks

Class Value for W1

A 1-126

B 128-191

C 192-223

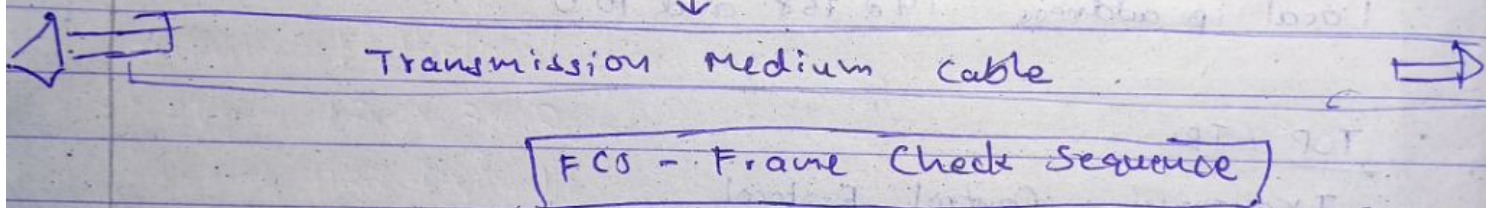
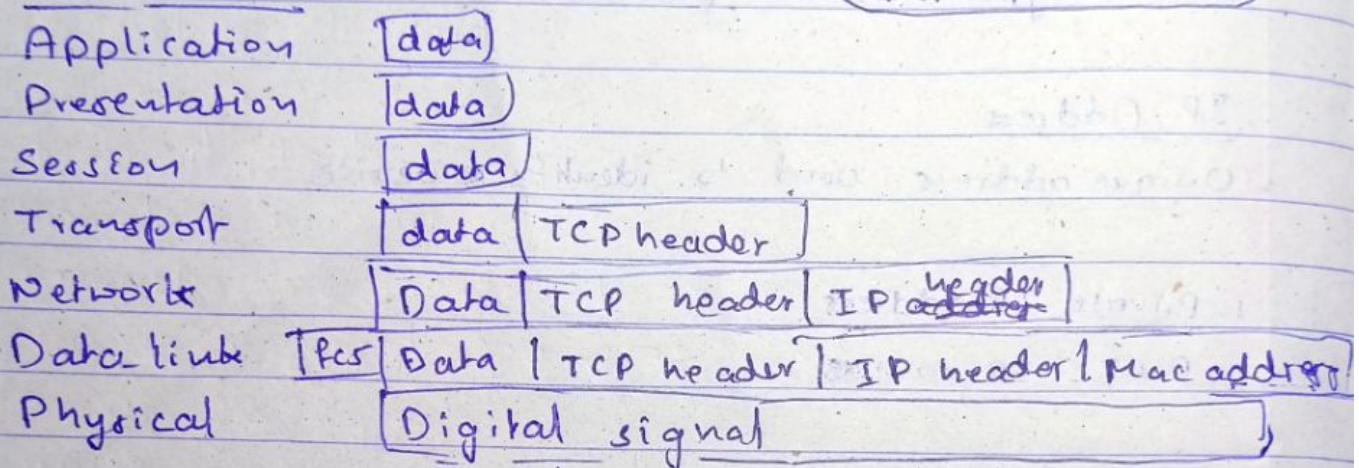
All People Seem To Need Data Processing

Subnet

Logical subdivision of an IP network.

192.168.1.0 / 24

OSI Model



Physical layer

- It is concerned with the transmission and reception of the unstructured raw bit stream over a physical medium.

Data Link layer

- Provides error-free transfer of data frames from one node to another over the physical layer.

Network Layer

- Controls the operation of the subnet, deciding which physical path the data should take based on network

Transport Layer

- Message segmentation : accepts a message from the layer above it, splits the message into smaller units (if not already small enough), and passes the smaller units down to the network layer

Session Layer

- The session layer allows session establishment between processes running on different stations.

Presentation layer

- Character code translation
- Data conversion ; bit order
- Data compression
- Data encryption

Application Layer

- The application layer serves as the window for users and application processes to access network services.

IEEE 802 Standards

Institute of Electrical and Electronic Engineers

Network Cabling

- Cable is medium through which information usually moves from one network device to another.

Types

- Twisted Pairs
- Shielded twisted pair (least expensive)
- Fiber Optic
- Patch cables
- Wireless Standards

Network Tools

Wireless network

- Enables connection to the network without using a wired connection.

IPv4

- Internet Protocol version 4
- Eg: 12.5.24.2
127.0.0.1
192.168.3.54
208.32.56.232

IPv6

- IPv6 is the new generation of IP addressing for the internet.

Port Number

- Port numbers primarily used to distinguish the application
- 0 - 65535

20221

FTP

23

Telnet

25

SMTP

53	DNS
80	World Wide web
110	POP3
144	News
6000	X- Windows

Basic commands

ping → Verify ip address exist
 ping site.com →
 ipconfig /renew /release → shows configurations
 tracert

Advanced TCP/IP commands

getmac
 arp -a
 nslookup site.com

All About DHCP

- Dynamic Host Configuration Protocol
- It is a client/server protocol that enables configured client computers to obtain IP addresses automatically.

DORA

- Discovery offer Request Acknowledgement
- DHCP utilizes ports 67 and 68

Remote Desktop Services

- RDS allows a user to take control of a remote computer or virtual machine over a network connection.

World Wide Web - Internet

Internet

- The internet is a worldwide system of connected computer networks.

World Wide Web

- WWW is an enormous system of interlinked hypertext documents that can be accessed by using a web browser.

Intranet

- Private computer network or single web site that an organization implements.

Extranet

- An extranet is similar to an intranet except that it is extended to users outside.

VPN

- Virtual Private Network
- Connection between two or more computers or devices that are not on the same private network.

Router

- A router is a device that forwards data packets along networks.
- A router is connected to two or more data lines from different networks.

Hub and Switcher

Hub

- A hub is the most basic central connecting device.
- Broadcast data to all devices.
- Receive & Broadcast

Switch

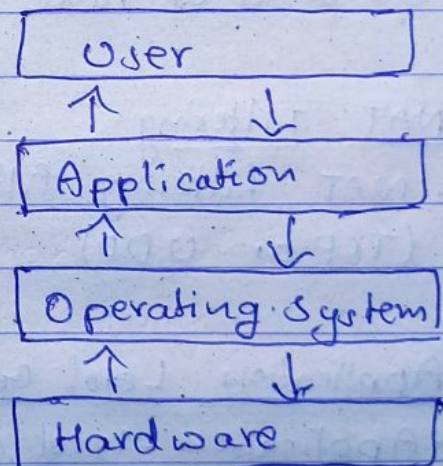
- Switches work the same way as a hub, but they can identify the intended recipient of the data.
- Can send and receive data at the same time.

OS Platforms

- An operating system is the most important software that runs on a computer.

Examples

- FreeBSD, NetBSD, OpenBSD
- Linux
- Microsoft Windows
- OpenVMS
- Mac OS
- OS/2



Network Intrusion

- A network intrusion is any unauthorized activity on a computer network.

Types of Attacks

- Traffic flooding
- Trojan
- Worms
- Brute force
- Denial of Service
- Port scan

Firewalls

- Firewalls are used to help protect a network from malicious attack and unwanted intrusion.

Packet filtering

- Packet filtering inspects each packet that passes through the firewall and accepts or rejects it based on a set of rules.

NAT Filtering

- NAT filtering filters traffic according to ports (TCP or UDP)

Application Level Gateway

- Application-level gateway (ALG) supports address and port translation and checks whether the type of application traffic is allowed.

Circuit-Level Gateway

- Circuit-level gateway works at the session layer of the OSI model when a TCP or UDP connection is established.